

Living Habitat

Rewilding & Architecture — Multi-Species Birdhouse & Living Roof

Inspired by Nordic angular roofs and green architecture, this birdhouse is designed to house multiple birds while acting as a foundation for plant and moss growth—blending into its garden surroundings over time.

Discipline

Product & Spatial

Context

CSM Rewilding Brief

Tools

Birch Plywood, Woodshop

Outcome

Full-scale structural birch plywood habitat built from hand-drafted trigonometric projections, iterative 1:5 paper nets, and 1:2 cardboard prototypes.

Research part 1 — architectural birdhouse precedents



Fig. 01 — 'Put A Bird In It' competition precedent



Fig. 02 — Klasern Nordic birdhouse



Fig. 03 — 'Spire' angular timber nesting box

Multi-Compartment Nesting & Angular

Analysing contemporary architectural birdhouses revealed how steep, intersecting Nordic gables can shelter multiple nesting chambers beneath a single roofline while creating natural valleys where moss and seedlings can take root.



Fig. 04 — 'Dwellings' birdhouse & sanctuary



Fig. 05 — Treehouse with birdhouses

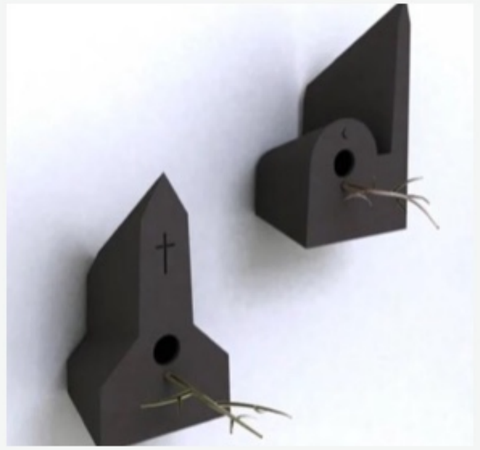


Fig. 06 — Frederik Roije: Holy Homes

Research part 2 — living roofs & wildlife habitats



Fig. 07 — Renzo Piano: California Academy of Sciences



Fig. 08 — 'Living Roof' Birdhouse



Fig. 09 — Tom Stuart-Smith: Knepp Walled Garden



Fig. 10 — Patrick Blanc: Vertical Garden



Fig. 11 — 'Moss Roof' Birdhouse



Fig. 12 — Miranda Brooks: Wildlife landscape design

Synthesising Architecture & Rewilding

Studying Renzo Piano and Patrick Blanc alongside British landscape designers Tom Stuart-Smith and Miranda Brooks showed that rough, untreated timber planes and recessed roof troughs encourage rapid colonisation by native mosses and sedum.

Initial ideation & trigonometric development

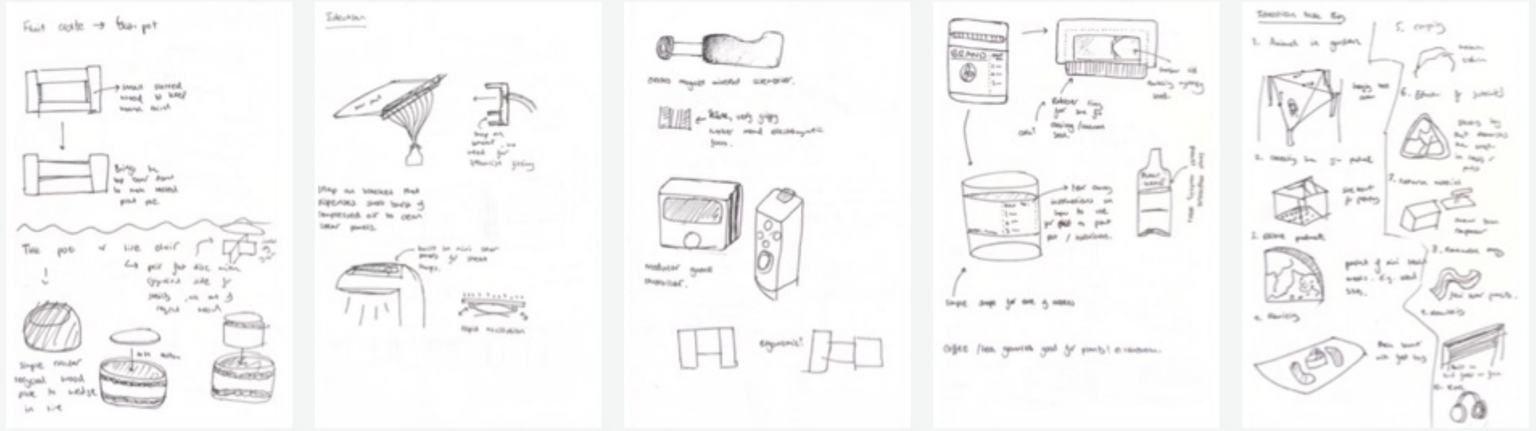


Fig. 13–17 — Initial breadth ideation: recycled crate bean-pot, solar street-lamp air bracket, modular controller, tear-away label, and 10-idea matrix

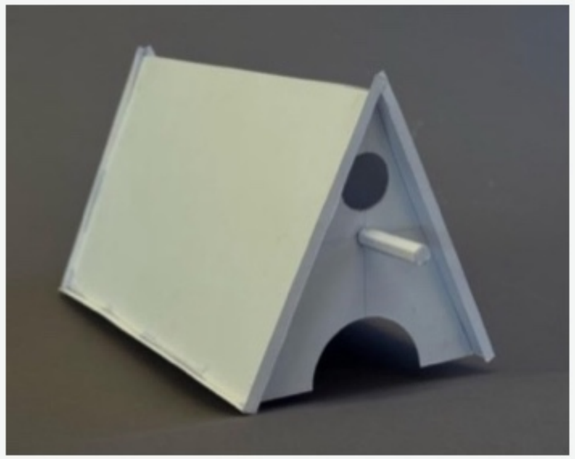


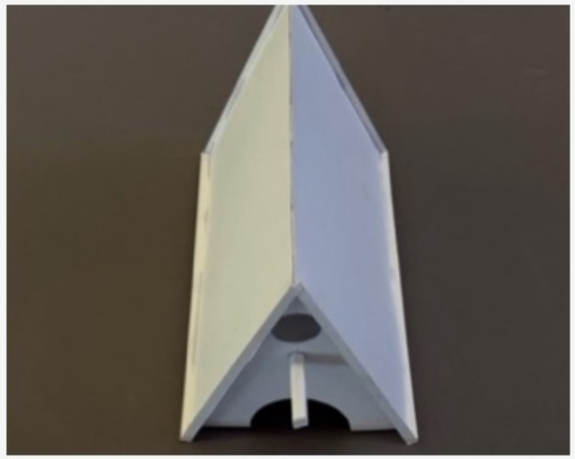
Fig. 18–20 — Developing the chosen birdhouse: orthographic elevations, roof pitch studies, and trigonometric dihedral angle calculations

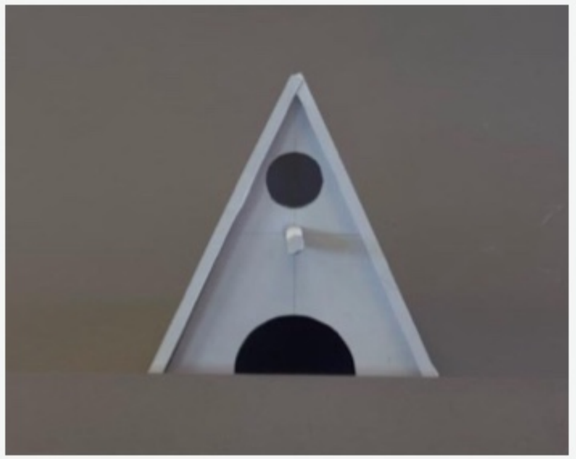
Calculating Compound Roof Angles by Hand

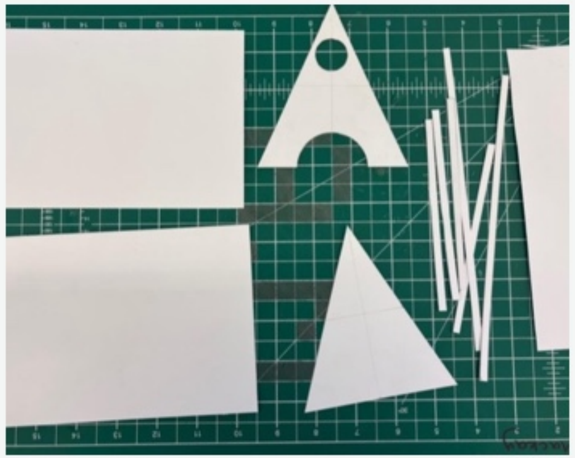
Because the central moss-trough valley is formed by two asymmetric gabled roofs meeting at a 45° plan angle, I used hand-drafted orthographic projections and trigonometry to calculate every true panel length and mitre bevel before cutting paper templates.

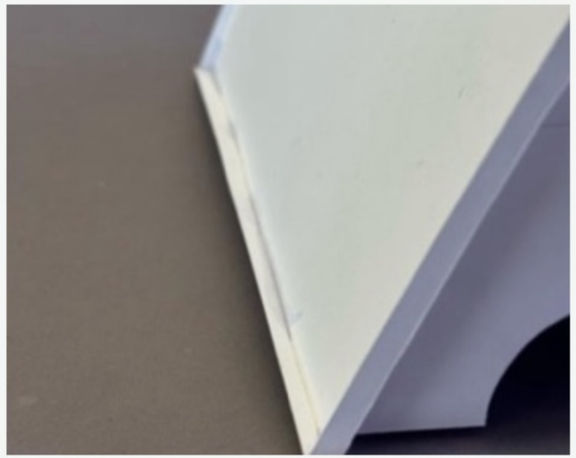
Small-scale prototype — master templates & 1:5 paper model

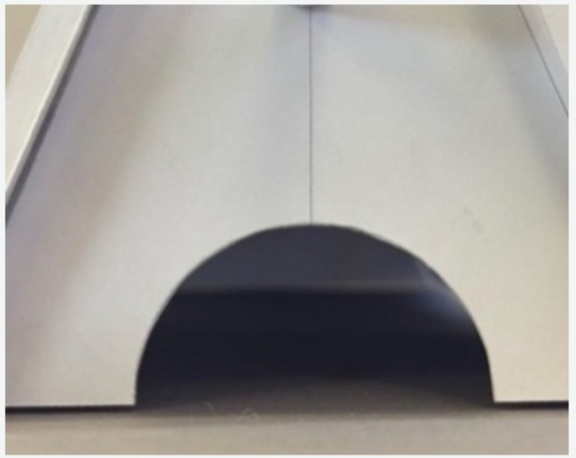












Master Template System

Each prototype was scaled directly from a single set of numbered master templates.

Building the 1:5 paper model confirmed that the compound valley angles closed cleanly without gaps and allowed me to test perch dowels and entrance hole proportions early.

Fig. 21–26 — Master numbered paper templates (1:5 scale) and taped paper net assembly verifying compound roof valley intersections

Scale prototypes — 1:2 cardboard & 1:1 timber test



Fig. 27–34 — 1:2 corrugated cardboard model (top) and 1:1 plywood test build (bottom), including outdoor garden placement trial

Accounting for Sheet Thickness

Moving from zero-thickness paper to 4mm corrugated cardboard and 9mm plywood exposed how sheet thickness offsets compound mitre seams.

Testing the 1:1 wooden prototype outdoors in a garden border validated the overhang depth and perch placement.

Technical drawings & workshop manufacturing

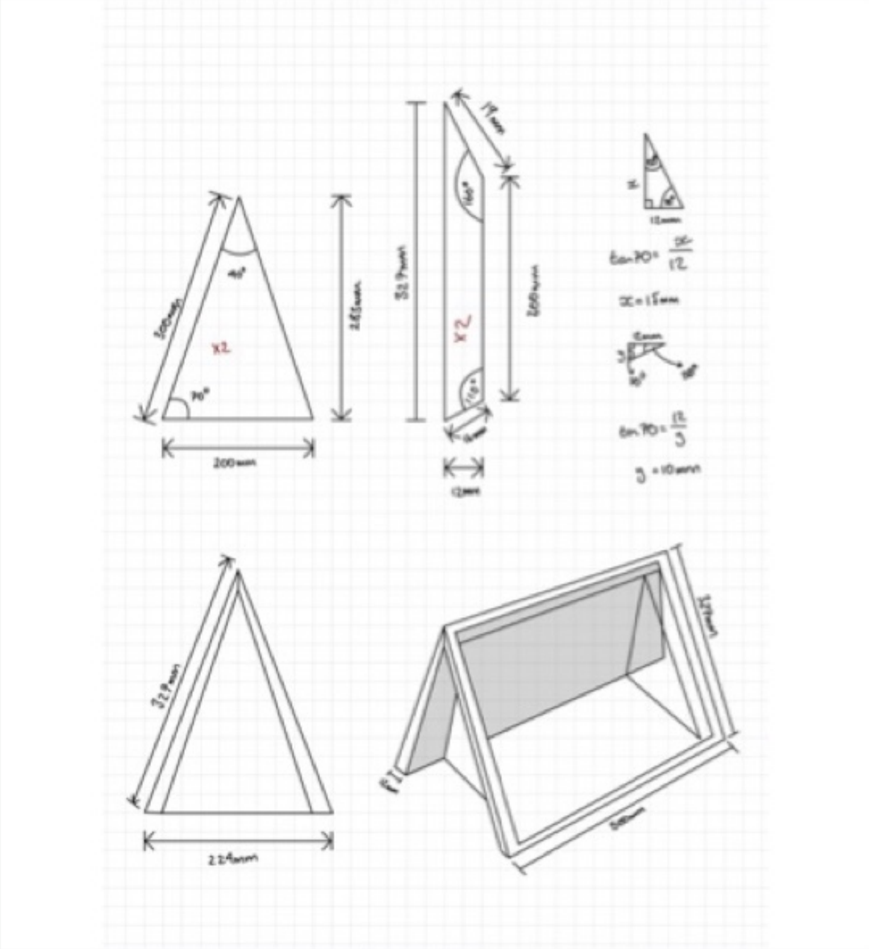


Fig. 35 — Final dimensioned orthographic technical drawing sheet (Front, Side, Plan & Roof Angles)

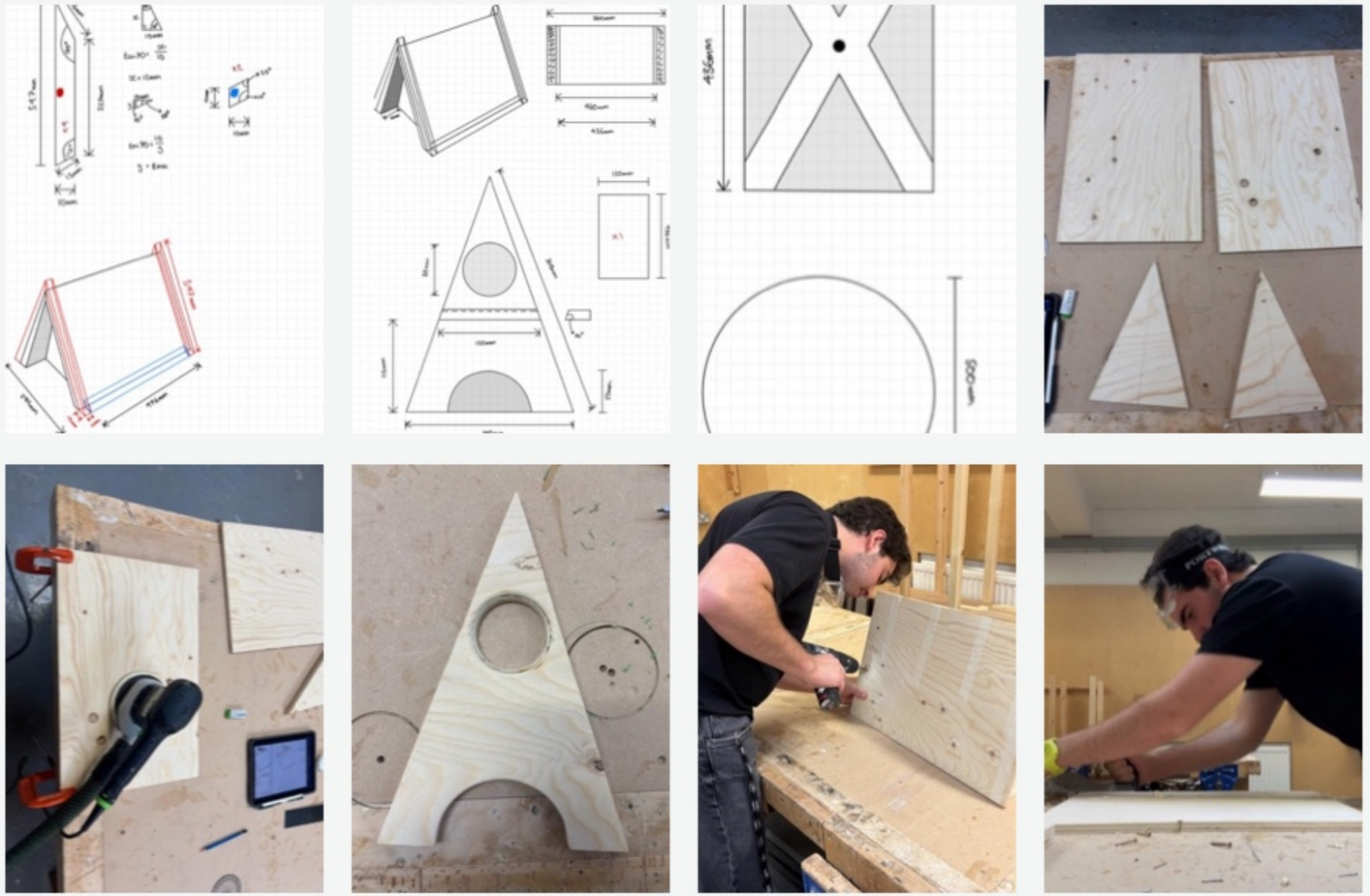


Fig. 36–43 — Workshop fabrication: marking birch plywood from templates, bandsaw bevelling, boring 32mm entrance holes, sanding, and gluing

Final product — birch plywood habitat





Fig. 44–45 — Completed birch plywood Living Habitat: front elevation and three-quarter detail showing central moss trough and dowel perches

Built to Weather & Rewild

Fabricated from 9mm exterior-grade birch plywood with precision bevelled roof joints and solid timber perches, the finished habitat provides two sheltered nesting cavities beneath a central living-roof valley.

01 **Dual Nesting Chambers** — Two independent 32mm entrance apertures sized for garden tits and sparrows.

02 **Living Moss Trough** — Central angled roof valley retains substrate and moisture for moss and sedum.

03 **Untreated Birch Exterior** — Weathers naturally to silver-grey, integrating into garden foliage over seasons.